MSc Data Analytics
Research Project Guidelines

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1. Introduction

The Research Project Module, AM41PR, is a compulsory, non-condonable[1] module which constitutes 60 credits of the 180 credits required for an MSc degree. It is not possible to obtain a full MSc degree without passing this module. Credits for this module will only be awarded after passing all taught modules in the programme.

[1] This means that being awarded the credits for this module requires the final marks to be strictly above or equal 50. Any marks less than 50 will require the student to retake the module.

This document gives vital information on carrying out your project. Please read it carefully and go back to it often in order to be aware of your responsibilities throughout the module.

All relevant deadlines will be communicated on the appropriate Blackboard page for your cohort.

2. The Research Project

Your project supervisor will provide guidance in all stages of your project work, but the responsibility for carrying out the work is yours alone.

Note that you are expected to show a capacity for independent work without excessive reliance on your supervisor. You should first attempt to resolve difficulties on your own initiative, e.g. by consulting manuals, discussion forums, online tutorials, etc. Please note that your supervisor will give you advice and feedback on how to carry out and manage your project, but cannot be expected, for instance, to do your calculations or write your code, should your project require it. To successfully manage your project, you should closely follow a project plan which identifies the tasks to be completed and the schedule for completing those tasks. A popular method for presenting a project plan, which is the standard on the industrial setting, is a **Gantt chart**. You are free to keep track of your project plan in any other suitable diagrammatic format if you feel appropriate.

You should meet your supervisor regularly throughout the project. The typical frequency is one meeting per week, but it It is vital to make an early start and work in a well-disciplined manner.

2.1. Supervision Arrangements

You should arrange to meet your project supervisor on a regular basis by setting up regular weekly meetings. Extra meetings may be needed and arranged near the beginning and end of the project or to resolve any problems that arise. Meetings may be brief if satisfactory progress is being made, but should not be skipped. You will find that obtaining regular feedback from your project supervisor on your progress and work done so far will help you improve the quality of your work.

Supervisory meetings are expected to take the form of one-to-one meetings unless agreed with your project supervisor otherwise. Video calls or virtual meetings can be arranged and are appropriate, especially if you are working on an industrial project at a company that is not located in Birmingham.

In the event of dissatisfaction regarding project supervision arrangements, you should try to discuss the issues with your project supervisor in the first instance. If the issues persist and/or the issues are of a nature that is inappropriate to discuss with your project supervisor, contact your Programme Director, the Head of Mathematics, or the Associate Dean of Postgraduate Programmes, whomever is more appropriate for the nature of the issues.

3. The Dissertation

The project work and outcomes must be carefully described, evaluated and presented for examination via a written dissertation by the appropriate date for your cohort (see table at the top of this document). Before starting to write your dissertation, you should agree its overall structure and contents with your supervisor (**see section 4.6**). Supervisors will be prepared to comment on the

general organisation and quality of presentation of a draft dissertation, provided they are given sufficient time, but cannot be expected to undertake detailed proof-reading.

The Learning Development Centre (LDC) at Aston University runs workshops and one-to-one tutorials on a range of topics, including academic writing and strategies for improving proof-reading skills. If English is not your first language, there may be help and resources available for you from the Centre for English Language and Communication at Aston (CELCA) at Aston University. You are encouraged to use the resources provided by the LDC[2] and/or CELCA[3] when preparing your dissertation.

3.1. Length

The dissertation should be concise and consistent with an adequate account of the work done. The core of the dissertation, which is composed of the main text excluding title, abstract, references and appendices, should comprise 10,000 words (with a tolerance of about 10% either below or above this limit) or a maximum of 40 pages, whichever is shorter. The total limit, including all other elements, should not exceed 60 pages.

If it seems likely that your dissertation will exceed the above limits or be well below the required amount of words, you need to discuss the appropriateness of the content with your supervisor. In case you exceed the 60-pages limit, you will need to obtain approval from the Programme Director to submit your dissertation. This request should include a justification of why the dissertation length cannot be reduced any further without compromising the presentation of the work and evidence that you have discussed and obtained the agreement of your supervisor.

3.2. Appendices

Every project submission must be a self-contained, complete record of the work done. Assessors must be provided with access to all data, calculations and computer code (written and commented version, not a computer file) such that they will have the means to check and test all your results.

These and any other materials necessary for a detailed evaluation of your work should be included in the appendices of your dissertation. If you are in doubt about what parts of your dissertation should comprise its core or the appendices, you should seek advice from your supervisor.

3.3. Writing style

The dissertation should be written in measured and scholarly English similar to that found in scientific journals. Bullet points may be used occasionally when appropriate (e.g., to emphasise a sequence of key issues), but should not be the main means of expression. Avoid using the first person singular ("I wrote . . .", "I designed . . .", etc.), except when expressing a personal opinion.

Clear, concise expression and a high standard of organisation and presentation will gain credit; opacity, verbosity, disorganisation, repetitiousness or an excessive number of spelling or grammatical errors will lead to a lower mark (see Marking Criteria, available on the MSc Data Analytics Blackboard page).

[2] See the section on "Essential Skills for Study" in LDC's website at: https://www2.aston.ac.uk/current-students/academic-support/ldc/essential-skills-for-study

[3] See CELCA's website at: https://www2.aston.ac.uk/clipp/english

3.4. Referencing

Any material in your report that comes from an external source is called a quotation. The nature of the material does not matter – whether it is English text, program code, diagrams, or a mixture, if it is taken from an external source, it is a quotation. The nature of the external source also does not matter – it may be a textbook, a website, a reference manual, a magazine, a radio broadcast, a video, etc. – if material in the report comes from anywhere except your own mind, it is a quotation. All quotations must be clearly identified as such. English text quotations must be enclosed in double quote marks (". . ."). Quotations of a paragraph or more should preferably be made to stand out better by indenting the text from the left margin and/or putting in the text in italics, as well as enclosing it in ". . .".

Due to the nature of mathematical knowledge, common sense must be used in some cases. You do not need to cite original papers when referring to a well-stablished mathematical theorem or method, although that might be needed with more recent ones. When in doubt, you should seek advice from your supervisor.

Near to (usually immediately after) a quotation of any kind, you must include a reference code which links to full details of the source of the quotation, which must appear in the **References** or **Bibliography** section of the dissertation. There are different system of reference and none of them is preferred as long as references are correctly made within your chosen system to the different kinds of documents. In particular, you should check the differences between referencing books, articles, online documents and any other additional kinds of sources.

For more information about how to manage your references and citations, and how to prepare quotations and references, see:

https://libguides.aston.ac.uk/c.php?g=99969&p=647394

3.5. Format

All dissertations must be prepared using a word processing software application for printing on A4 paper. The preferred application is LaTeX.

Unless otherwise agreed with your supervisor, the descriptive text of the dissertation should be in a 12 point font, preferably Arial or Times (other fonts and sizes may be used for headings, footnotes, etc.).

Descriptive text should use 1.5 line spacing.

Margins should be at least 35mm on the binding edge and 15mm elsewhere.

Pages must be numbered. Chapters and sections should also be numbered.

The core text of the dissertation and any appendices must constitute a single document in PDF format.

There must be a title page showing the title of the dissertation, the full name and the student number of the candidate, the degree for which the report is submitted, and the year and month of submission.

Failure to adhere to requirements stated above may result in loss of marks.

3.6. Structure and Contents

You should discuss the organisation of your dissertation with your project supervisor. The following structure provides a starting point, however, modifications may be needed for your project and

variations are allowed as long as they do not affect the clarity of the dissertation and all necessary elements are present at some point.

If you undertake a project with any significant business element, e.g. a project with a real-world business client, you should include in-depth consideration of the business context and business related evaluation.

- **Title Page:** should contain the title of the dissertation and the name of the student and their supervisor.
- **Abstract:** it is a brief description of the project, its aims and conclusions, in a separate page right after the title page.
- **Acknowledgements (Optional)**
- **List of Contents**
- **Introduction:** this is the first chapter of your thesis, usually containing a summarised description of the motivation, benefits, aims and applications. The introduction normally do not contain any technical details, except for superficial ones.
- **Context/Literature Review:** it should include an up-to-date description of the importance of the project, what has been done and the current research in the area. Applications, challenges and any other considerations about the project's suitability within the appropriate should be included with all relevant references.
- **Problem Description:** describe the problem both at high level and in detail, highlighting any identified issues and showing why the problem is important and should be tackled. Describe any work needed to establish detailed project requirements, ending with a clear statement of the requirements that the project's end- product should meet. Requirements should be logically derived from the aims of the project, not merely asserted. Different projects may have different types of end-products.
- **Theory and Methods:** because the dissertation should be self-contained, a detailed description of all theory and methods used should be included in the core text. Describe how you solved the problem using appropriate examples and diagrams in sufficient detail to allow the project work to be checked and reproduced by third parties.
- **Evaluation:** realistic evaluation is a vital part of a good project report. Describe evaluation methods; discuss reliability and usability of the end-product. To what extent have the specified objectives of the project been met? How successful and appropriate was the methodology adopted? What is the business value of the end-product if relevant? What is the value of the contribution made by the project to the appropriate knowledge area?

  The evaluation should also include a discussion on the ethical implications and the potential practical value (business/social/health/scientific) to suitable stakeholders where relevant. An example of project ethics to discuss would be: "What impacts machine learning would have to society if used to assess medical data?" If you believe there are no ethical implications of your work, that should clearly be stated in your evaluation with a sound justification.

- **Conclusion:** briefly restate the work done; summarise any findings or recommendations emerging from the project; acknowledge any limitations of the work and/or give any suggestions for further work.
- **References:** use a recognised system consistently. List all references together at the end of the dissertation, not after each chapter. Links to webpages should be in this section.
- **Appendices:** the exact contents will depend on the nature of the project, but in general should include all detailed calculations and computer code relevant to the project. A reader of the report

should be able to reproduce all results by referring to this information. Other items that are likely to be appropriate include: example results, evidence of test procedures and outcomes, questionnaires, summary notes and/or transcripts of interviews, a collection of demonstration screen shots too large for the main body of the dissertation, ethics approval (if applicable).

3.7 Data

Although data are suggested to be included in the appendices, there might be situations in which datasets are too large for this to be feasible. In such cases, we recommend that all data deemed relevant by you and your supervisor should be stored in an online repository of your own choice and due access should be granted to all examiners. A link to the dataset should be include in the references and cited whenever appropriate in the main text.

Currently, there are no official data repositories at Aston University. Therefore, you are free to choose one that suits your needs as long as it is in agreement with your supervisor.

4. Submission of the Dissertation

Your final MSc dissertation should be submitted electronically on Blackboard. You will be notified of the details by email and through an announcement on Blackboard before the advertised deadline. It is expected that the submission will consist of one single PDF file. Any code and supplementary material that you produce for the MSc project should be presented in the appendices to your thesis in the same file. For large datasets, see **section 3.7**.

4.1. Lateness

Section 5 of the University's Assessment Regulations 2020-21 (AU-RSC-19-2814-D) states:

"The late submission penalty will be 10% of the awarded mark for each working day that the piece of work was submitted after the formal deadline."

Waivers of lateness penalties may be granted in exceptional circumstances, but ONLY on written submission to the School's Exceptional Circumstance Assessment Panel (ECAP)[4]. Project supervisors are NOT authorised to grant extensions to the project submission deadline. Any request for an extension must be given in writing to the Programme Director who, where appropriate, will forward it to the Associate Dean for Postgraduate Studies for approval. Any such request should be backed up by evidence that your ability to work on the project was affected for some period of time. Requests for extensions of more than one month must be accompanied with an exceptional circumstances case.

It is extremely unlikely that multiple extensions will be granted, so requests for extensions should normally be submitted towards the end of the project, although the Programme Director should be informed of your intention to request an extension as soon as possible.

[4] See https://www2.aston.ac.uk/clipp/quality/a-z/exceptional-circumstances for details on how to prepare an exceptional circumstance case for submission.

5. Assessment

Your Research Project will be assessed by two examiners. The first examiner will be either your supervisor or a member of the supervisory team appointed by them. The second examiner should be someone who has not been involved in the supervision and will also be appointed by your supervisor. Your final marks will be agreed by both examiners. The detailed marking scheme is available on the MSc Data Analytics Blackboard page and at the end of this document.

After submission, a copy of your written MSc dissertation will be sent to each of your examiners to be assessed according to the provided marking criteria. The final marks and any due feedback will be released after consideration by the exams board.

5.1. Marking Criteria

There are 4 main elements composing the marking criteria:

- **Knowledge and understanding of the project context and foundations:** 20%
- **Technical challenge and achievements for a 3-months project at MSc level:** 40%
- **Technical writing and presentation:** 20%
- **Evaluation and conclusions:** 20%

A detailed description of the requirements for each of these elements and the corresponding awarded marks are given in the tables below.

Element (Weight)	Range (%)	Description
Knowledge and understanding of the project context and foundations. (20%).	100 – 86	Outstanding focused coverage of the topic's background and justification of the techniques. Outstanding review of high-quality publications. Evidence of deep understanding of the background and the review content.
	85 – 68	Excellent focused coverage of the topic's background and justification of the techniques. Excellent review of related works. Indication of understanding of the review content.
	67 – 58	Solid focused background. The sections read well and demonstrate having collected relevant references on the project's topic. Some justifications of the techniques are given.
	57 – 48	Background of acceptable quality. The coverage and literature review read well, but do not dive deep enough into the topic.
	47 – 40	Unacceptable quality. Insufficient coverage of the background. Involves incoherences, shows inaccuracies and evidence of lack of understanding.
	39 – 1	Poor quality. Major incoherences and evidence of serious lack of understanding of the project's background.

Element (Weight)	Range (%)	Description
Technical challenge and achievements for a 3-months project at MSc level. (40%)	100 – 86	Outstanding delivery, which could be acceptable as the core of a publication. Evidence of impressive creativity, independence and originality. Evidence of full awareness of the research context. Extensive comparisons with the state of the art are accurately reported.
	85 – 68	Excellent and well-rounded delivery in all the aspects. The project achievements show high degree of creativity. Numerical tests, where appropriate, are comprehensive. Some comparisons with the state of the art are produced.
	67 – 58	Solid and convincing delivery, which shows clear level of competence and some level of creativity. The size and complexity of the tasks are substantial within a 3-months project. There may be some questionable ways of tackling non-central aspects of the project, and numerical tests, where appropriate, might have some weakness.
	57 – 48	The delivery involves modest level of complexity and creativity. The numerical tests, where appropriate, may be minimally convincing. Some aspects of the project may be questionable. The numerical tests, where appropriate, may be incomplete or superficial.
	47 – 40	The delivery of the projects is not acceptable at MSc level. The tasks do not represent a complex enough challenge. The numerical tests, where appropriate, may not be correctly designed to draw conclusions.
	39 – 1	Poor or no delivery at all. No evidence of having achieved the project goals. The student may have done some incoherent work but there may not be evidence of having learned anything related to the project.

Element (Weight)	Range (%)	Description
Technical writing and presentation. (20%)	100 – 86	Outstanding presentation of the project in all its aspects, including creativity and originality. At this level the technical writing achieves what is expected to publish the work in a high-quality journal.
	85 – 68	Excellent presentation. No concept is used without previous accurate definition. Notation is throughout coherent.
	67 – 58	Good technical presentation. Most concepts are well defined in mathematics terms, but there may be some imprecision or room for improvements.
	57 – 48	It is clear that there is an effort dedicated to the technical presentation of the material, but some definitions are not given with acceptable precision. There may be some incoherence in notation or technical flaws.
	47 – 40	The technical presentation is not acceptable. Most concepts are not defined with enough precision. The quality of figures and tables may be poor or provide little information. The references may not be reliable.
	39 – 1	Poor technical presentation in all its aspects.

Element (Weight)	Range (%)	Description
Evaluation and conclusions. (20%)	100 – 86	Definite conclusions are drawn on the basis of overwhelming evidence produced within the project. The evaluation of results involves comparisons with the state of the art and recent publications.
	85 – 68	Strong and clear conclusions drawn from enough evidence. The evaluation involves comparison with techniques that might be considered traditionally state of the art in 'some sense' but that might not be recent.
	67 – 58	Solid conclusions that may not be definite due to lack of complete evidence. The evaluation involves some comparisons with related works.
	57 – 48	The conclusions are acceptable, but more evidence should have been collected to draw more reliable conclusions. The evaluation may not involve comparisons with other reliable related work.
	47 – 40	Unacceptable conclusions based on no evidence collected within the project. The evaluation is poor and do not include any sort of comparison with other works.
	39 – 1	Poor or wrong conclusions. Poor, incoherent or wrong evaluation of the project outcomes.

6. Originality

The dissertation and the material it is based on must be entirely your own work, except where the report clearly states otherwise. As indicated earlier, any report elements drawn from an external source must be clearly identified as quotations. In addition, if the practical work depends, to any significant extent, on someone else's work or previous work of your own (e.g. undergraduate project, coursework or placement work), the source of the material relied on and the extent of the dependence must be clearly stated in the report. For a project to rely on previous work is NOT a deficiency. There is no benefit in "reinventing the wheel" – indeed it is meritorious to make good use of previous work by yourself or others – but what is essential is that all work relied on is clearly and accurately acknowledged.

Claiming someone else's work as your own in a project report is as much cheating as is copying in a written examination and will be treated similarly by the University's disciplinary procedures. It is also an offence to reuse other work of your own, e.g. from other modules or other activities you undertake inside/outside the university, without acknowledging its origin.

6.1. Plagiarism

Making use of the intellectual work of others without properly acknowledging the origin of the work is the academic offence known as **plagiarism**. As well as actual plagiarism, any action which suggests an intention to plagiarise can get you into trouble. In particular, you must not post anything to do with your project work on a website outside the University without written permission. You must not look for other people to do work for you, whether or not for payment. All such offences will be subject to the University's disciplinary procedures which may lead to expulsion.

The LDC hosts study guides and runs workshops and one-to-one tutorials on referencing and avoiding plagiarism. You are encouraged to use the resources provided by the LDC when preparing your dissertation.

The Library Services hosts information and guidelines on referencing as well as copyright and intellectual property. You are encouraged to visit these resources when conducting your project work at:

https://libguides.aston.ac.uk/Copyright

6.2. Formal rule

It is each student's individual responsibility to be aware of the information in this document and in all applicable University regulations.

6.3. Sanctions

Project reports or practical work that attract suspicion of plagiarism or collusion will be reported for investigation under the University's disciplinary regulations. The penalties that can be imposed if an offence is found to have occurred include a zero mark for the entire project and in very serious cases expulsion from the University.

Marks may be lost for academic deficiencies that appear to fall short of deliberate plagiarism, e.g. a poor standard of referencing. Please take careful note of the advice on quotations and referencing in **section 3.4** of this document.

7. Useful Links

Aston University (2020).Aston University Assessment Regulations 2020-21 (AU-RSC-19- 2814-D) [online]. Available at https://www2.aston.ac.uk/clipp/documents/Quality/

Regulations/202021/used/final-au-rsc-19-2814-b-assessment-regulations-2020-21. pdf [Accessed 13th January 2021].

Library Information Services (2018). LibGuides: Copyright and Intellectual Property [online]. Available at http://libguides.aston.ac.uk/Copyright [Accessed 13th January 2021].

Library Information Services (2021). LibGuides: Mathematics Referencing [online]. Avail- able at https://libguides.aston.ac.uk/c.php?g=99969&p=647394 [Accessed 1st June 2021].

Springer Link (2021). Ethics and Information Technology [online]. Available at https:// link.springer.com/journal/10676 [Accessed 13th January 2021].